

Grade 2 Ontario Curriculum Algebra

If-Then Detectives

Unit Overview: If-Then Detectives

Grade: Grade 2 | **Strand:** Algebra

Target Expectations: C3.1, C3.2

Learning Goal: Children will write and execute code with conditional statements; read and alter existing IF-THEN code; and describe how changes to the conditions affect the outcomes.

Lesson	Lesson Title	Learning Goal
Day 1	What's an IF-THEN?	I can read an IF-THEN statement and explain what it means.
Day 2	Writing IF-THEN Code	I can write IF-THEN code for a math situation.
Day 3	Reading IF-THEN Code	I can read IF-THEN code and predict the outcome.
Day 4	Altering IF-THEN Code	I can alter IF-THEN code and describe the effect.
Day 5	My IF-THEN Project Capstone	I can write my own IF-THEN code for a math task.

Lesson 1: What's an IF-THEN?

Goal: An IF-THEN statement says: IF a condition is true, THEN do an action.

Expectation: C3.1

Learning Goal: "I can read an IF-THEN statement and explain what it means."

Materials: M1_if_then_cards, M2_condition_cards, M3_action_cards, M7_sorting_mat

1. Minds On (12 mins): Inspector If's First Case

Inspector If says: IF it rains, THEN take an umbrella. What does that mean?
Inspector If opens: 'CHECK the IF — THEN do the THEN!' (Class echoes.)

Bring children to the carpet. Inspector If + Then the Action Dog puppets.
'Friends — Inspector If is a Conditional Cat. She lives by IF-THEN rules. Listen to her first rule.'

2. Action (25 mins): IF-THEN Sorting Stations

- Whole group: read 4 IF-THEN statements together.
- Send pairs to three stations (~7 min each).
- Stretch challenge at Station C: write an IF-THEN with a math condition (e.g., 'IF the number is greater than 5, THEN add 1').
- Discuss the big idea.
- Circulate to record formative evidence on the IF-THEN tracker.
- STRATEGY COMPARE: Two children share their invented IF-THENS — one with a math condition (IF >5 THEN clap) and one with a real-life condition (IF wearing red THEN stand).

3. Consolidation (13 mins): Inspector If's IF-THEN Brag

Discuss:

- What is the IF part called?
- What does TRUE mean?
- Can you give an IF-THEN example from your life?
- What happens if the condition is FALSE?
- Inspector If's preview: tomorrow Then the Action Dog helps us WRITE our own IF-THEN code (Lesson 2 — C3.1).

Exit: Inspector If's refrain — class chants 3x: 'CHECK the IF — THEN do the THEN!' Each child writes ONE IF-THEN they used today on a sticky.

Lesson 2: Writing IF-THEN Code

Goal: We can write IF-THEN code for math: IF the number is even, THEN clap.

Expectation: C3.1

Learning Goal: "I can write IF-THEN code for a math situation."

Materials: M1_if_then_cards, M2_condition_cards, M3_action_cards, M7_sorting_mat

1. Minds On (12 mins): Inspector If Writes Code

Inspector If wants to sort numbers. EVEN numbers — clap. ODD numbers — stomp. Help her write the code.

Bring children to the carpet. Inspector If + Then the Action Dog puppets. 'Friends — yesterday we READ IF-THEN code. Today we WRITE it.'

2. Action (25 mins): IF-THEN Writing Stations

- Whole group: write IF-THEN code together for sorting tasks.
- Send pairs to three stations (~7 min each).
- Stretch challenge at Station C: write a 3-branch IF-THEN-ELSE-IF (e.g., IF >20 say BIG; ELSE IF >10 say MEDIUM; ELSE say SMALL).
- Discuss the BRANCH idea.
- Circulate to record formative evidence on the writing tracker.

3. Consolidation (13 mins): Inspector If's Writing Brag

Discuss:

- What's a 'branch' in IF-THEN code?
- Why do we write TWO branches (TRUE and FALSE)?
- Can you give a math condition you used?
- What happens if you forget the FALSE branch?

Exit: Each child writes ONE two-branch IF-THEN on a sticky note: 'IF ____, THEN ____.'

Lesson 3: Reading IF-THEN Code

Goal: Reading IF-THEN code means following the condition and predicting the action.

Expectation: C3.2

Learning Goal: "I can read IF-THEN code and predict the outcome."

Materials: M1_if_then_cards, M2_condition_cards, M3_action_cards, M4_input_cards

1. Minds On (12 mins): Inspector If Reads Code

Inspector If's code: IF the number > 10, THEN say BIG. ELSE say SMALL. Trace it with input 7. What does it say?

Bring children to the carpet. Inspector If + Then the Action Dog puppets.
'Friends — yesterday we WROTE IF-THEN code. Today we READ it.'

2. Action (25 mins): IF-THEN Reading Stations

- Whole group: trace 4 IF-THEN code samples together.
- Send pairs to three stations (~7 min each).
- Stretch challenge at Station C: trace a 3-branch code (IF >20 BIG; ELSE IF >10 MEDIUM; ELSE SMALL) with mixed inputs.
- Discuss the trace process.
- Circulate to record formative evidence on the reading tracker.

3. Consolidation (13 mins): Inspector If's Reading Brag

Discuss:

- What does TRACE mean?
- What's the difference between condition and outcome?
- Why do we substitute the input first?
- Can two inputs give the same outcome?

Exit: Each child traces ONE IF-THEN code with a chosen input on a sticky note:
'Input: ____.

Lesson 4: Altering IF-THEN Code

Goal: Changing the condition or the action changes the outcome.

Expectation: C3.2

Learning Goal: "I can alter IF-THEN code and describe the effect."

Materials: M1_if_then_cards, M2_condition_cards, M3_action_cards, M4_input_cards

1. Minds On (12 mins): Inspector If Alters Code

Original code: IF >10, THEN BIG. New code: IF >20, THEN BIG. What changed?
Inspector If opens: 'NEW code, NEW trace!' (Class echoes.)

Bring children to the carpet. Inspector If + Then the Action Dog puppets.
'Friends — yesterday we READ code. Today we ALTER it. Altering means CHANGING.'

2. Action (25 mins): IF-THEN Alteration Stations

- Whole group: alter 4 codes together.
- Send pairs to three stations (~7 min each).
- Stretch challenge at Station C: chain TWO alterations on the same code.
- Discuss alteration types.
- Circulate to record formative evidence on the altering tracker.
- INVENT-AND-COMPARE: Each pair writes TWO different alterations of the same original code that produce the SAME altered outcome for a chosen input.

3. Consolidation (13 mins): Inspector If's Alter Brag

Discuss:

- What does ALTER mean?
- What 3 kinds of alterations did we try?
- Can an alteration leave the outcome the same?
- Why is altering useful?

Exit: Inspector If's Day-4 refrain — class chants: 'NEW code, NEW trace!' (3x).

Lesson 5: My IF-THEN Project Capstone

Goal: We can write our own IF-THEN code for a math task in our life. Sort, classify, decide, repeat.

Expectation: C3.1, C3.2

Learning Goal: "I can write my own IF-THEN code for a math task."

Materials: M2_condition_cards, M3_action_cards, M4_input_cards, M6_capstone_template

1. Minds On (12 mins): Inspector If's Final Showcase

Now you write YOUR IF-THEN code. Pick a math task. Write the code. Trace it.

Bring children to the carpet. Inspector If + Then the Action Dog together.
'Friends — today you become CODERS. Pick a math task. Write IF-THEN code. Trace it.'

2. Action (25 mins): Capstone IF-THEN Project

- Hand out capstone templates.
- Children trace their code with 3 sample inputs and predict outcomes.
- Stretch challenge: alter ONE part of the code (condition OR action) and predict the new outcome for the same 3 inputs.
- Gallery walk: children swap templates and trace each other's code with 1 new input.
- Whole-class share: 5-6 children share their code aloud.

3. Consolidation (13 mins): Inspector If's Final Brag and Certificate Ceremony

Discuss:

- What's the most surprising IF-THEN you saw today?
- Why do we write BOTH branches?
- What did you alter in your code?
- What's something in your life that uses IF-THEN logic?

Exit: Each child receives a Junior Conditional Coder certificate.

Name: _____

Date: _____

Inspector If's Cases

Learning Goal: I can read an IF-THEN statement and explain what it means.

Mission

1. Underline the IF (condition) and circle the THEN (action) in each statement.
2. For each IF-THEN with the given input, decide if the condition is TRUE or FALSE.
3. Write 3 of your own IF-THEN statements.

Truth Chart

Condition	TRUE	FALSE
Number > 5	☐	☐
Even number	☐	☐
Has wheels	☐	☐
Is red	☐	☐

Date: _____

Learning Goal: I can write IF-THEN code for a math situation.

1. Match each condition with an action that fits.
2. Write IF-THEN code with TWO branches for each sorting task.
3. Write 2 of your own two-branch IF-THENS.

Write IF-THEN code with TWO branches for each sorting task.

[illegible]

Date: _____

Learning Goal: I can read IF-THEN code and predict the outcome.

1. Trace each IF-THEN code with the given input.
2. For each IF-THEN code, list 2 inputs that give TRUE outcomes and 2 inputs that give FALSE outcomes.
3. Pick a code from above.

[illegible]

Name: _____ Date: _____

Name: _____ Date: _____

Inspector If's Alterations

Learning Goal: I can alter IF-THEN code and describe the effect.

Instructions for the Student:

1. Change the condition in each code.
2. Change the action in each code.
3. Swap the TRUE and FALSE branches in 2 codes.

Part 3: Swap Branches

Swap the TRUE and FALSE branches in 2 codes. Describe the effect.

[illegible]

Date: _____

Learning Goal: I can write my own IF-THEN code for a math task.

1. Pick a math task.
2. Pick 3 sample inputs.
3. Alter ONE part of your code.

[illegible]

IF-THEN Statement Cards

(M1_if_then_cards)

24 cards each showing an IF-THEN statement (everyday + math).

IF-THEN Cards

IF it rains THEN open umbrella	IF dark THEN turn on light
IF cold THEN wear coat	IF tired THEN sleep
IF hungry THEN eat	IF thirsty THEN drink
IF dirty THEN wash	IF late THEN run

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 4

Laminate: yes

USE IN CLASS

Quantity: 24 cards

Lessons: 1, 2, 3

Prep: ~20 min

TEACHER PREP

1. Print 4 pages on cardstock.

2. Laminate.

3. Cut into 24 cards.

4. Sort by complexity.

5. Distribute Lessons 1-3.

Condition Cards

(M2_condition_cards)

12 condition cards (e.g., 'IF the number is even', 'IF it is sunny').

Condition Cards

it rains	it's dark	you're cold	you're tired
you're hungry	it's late	you're thirsty	the door is locked

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

USE IN CLASS

Quantity: 12 cards

Lessons: 1, 2, 3, 4

Prep: ~12 min

TEACHER PREP

1. Print 2 pages on cardstock.

2. Laminate.

3. Cut into 12 cards.

4. Sort by type.

5. Distribute Lessons 1-4.

Action Cards

(M3_action_cards)

12 action cards (e.g., 'THEN clap', 'THEN take an umbrella').

Action Cards

open umbrella	turn on light	wear coat	go to bed
eat lunch	run faster	drink water	use the key

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

USE IN CLASS

Quantity: 12 cards

Lessons: 1, 2, 3, 4

Prep: ~12 min

TEACHER PREP

1. Print 2 pages on cardstock.

2. Laminate.

3. Cut into 12 cards.

4. Sort by type.

5. Distribute Lessons 1-4.

Input Number Cards

(M4_input_cards)

20 input number cards (numbers 1-50) for tracing IF-THEN code.

Input Cards

7	12	even	odd
red	blue	BIG	small
yes	no	5+5	3-1

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 2

Laminate: yes

USE IN CLASS

Quantity: 20 cards

Lessons: 3, 4

Prep: ~12 min

TEACHER PREP

1. Print 2 pages on cardstock.

2. Laminate.

3. Cut into 20 cards.

4. Sort by value.

5. Distribute Lessons 3-4.

Code Alteration Cards

(M5_alteration_cards)

8 cards each with an original IF-THEN and an alteration instruction.

Alteration Cards

change condition	swap action
add an ELSE	remove a step
add a check	use AND
use OR	wrap in loop

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 8 cards

Lessons: 4

Prep: ~10 min

TEACHER PREP

1. Print 1 page on cardstock.

2. Laminate.

3. Cut into 8 cards.

4. Sort by alteration type.

5. Distribute Lesson 4.

Capstone IF-THEN Template

(M6_capstone_template)

Letter-size template for Lesson 5 capstone.

Capstone IF-THEN Template	
1	Write Code _____
2	Trace _____
3	Alter _____
4	Re-Trace _____

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: no

USE IN CLASS

Quantity: 1 per child

Lessons: 5

Prep: ~12 min

TEACHER PREP

1. Print 1 per child.

2. Hand out Lesson 5.

3. Children fill in.

4. Collect for assessment.

5. Display gallery.

IF-THEN Sorting Mat

(M7_sorting_mat)

Letter-size mat with TRUE column and FALSE column.

Sort Mat: Conditions vs Actions

CONDITIONS (IF...)	ACTIONS (THEN...)

PRINT SPECS

Size: letter landscape
Colour: bw
Pages per set: 1
Laminate: yes

USE IN CLASS

Quantity: 1 per pair
Lessons: 1, 2, 3
Prep: ~10 min

TEACHER PREP

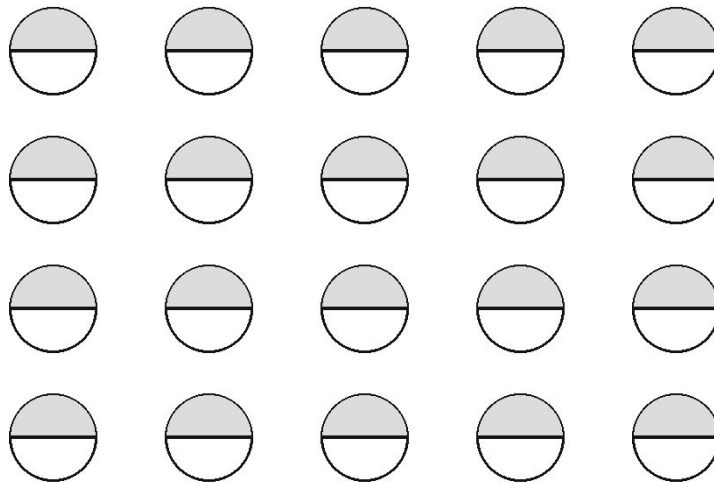
1. Print 1 per pair.
2. Laminate.
3. Hand out Lesson 1.
4. Use dry-erase.
5. Reuse Lessons 2-3.

Two-Colour Counters

(M8_two_colour_counters)

20 counters per pair for IF-THEN tracing.

Two-Colour Counters (cut & flip)



TRUE = light side up FALSE = dark side up

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 20 per pair

Lessons: 2, 3, 4

Prep: ~15 min

TEACHER PREP

1. Print 1 sheet per pair.
2. Laminate.
3. Cut into 20 counters.
4. Sort into pair bags.
5. Distribute Lessons 2-4.

TRUE / FALSE Anchor Chart

(M9_truth_chart)

Large classroom anchor chart explaining TRUE and FALSE.

Truth Chart

Condition	TRUE	FALSE
Number > 5	☐	☐
Even number	☐	☐
Has wheels	☐	☐
Is red	☐	☐

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 4

Laminate: yes

USE IN CLASS

Quantity: 1 chart

Lessons: 1, 2, 3, 4

Prep: ~15 min

TEACHER PREP

1. Print 4 pages and tile.

2. Laminate.

3. Tape to wall Day 1.

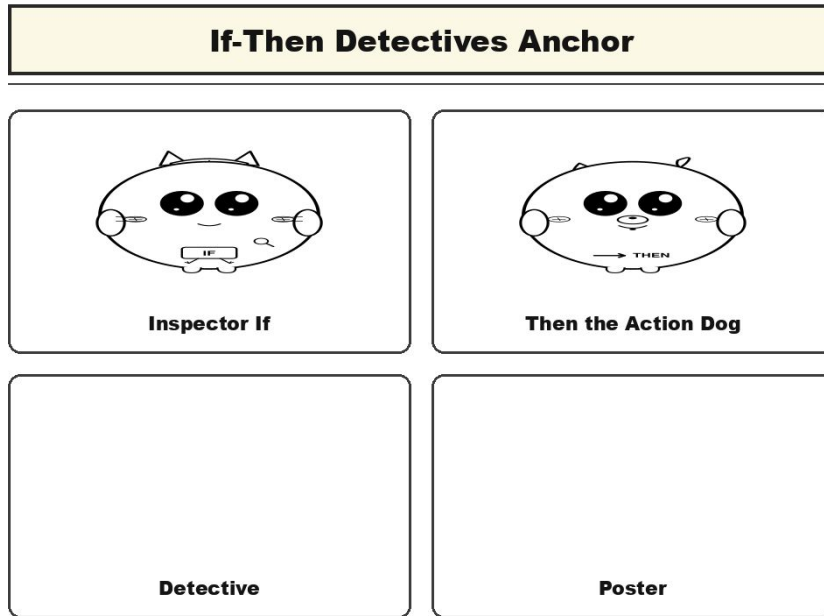
4. Reference all week.

5. Roll for re-use.

If-Then Detectives Anchor

(M10_if_then_poster)

Large classroom anchor showing Inspector If and Then the Action Dog with sample IF-THEN code.



PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 4

Laminate: yes

USE IN CLASS

Quantity: 1 poster

Lessons: 1, 2, 3, 4, 5

Prep: ~18 min

TEACHER PREP

1. Print 4 pages and tile.

2. Laminate.

3. Tape to wall Day 1.

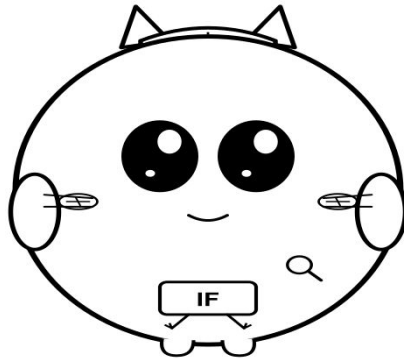
4. Reference all week.

5. Roll for re-use.

Inspector If Character Puppet

(char_if_puppet)

Printable Inspector If illustration on a popsicle stick.



Inspector If the Cat

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

TEACHER PREP

1. Print on cardstock.

2. Cut.

3. Glue popsicle stick.

4. Optional: laminate.

5. Store flat.

USE IN CLASS

Quantity: 1 puppet

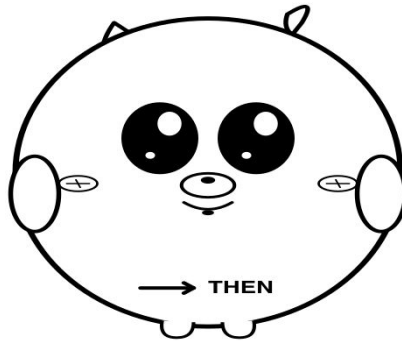
Lessons: 1, 2, 3, 4, 5

Prep: ~10 min

Then the Action Dog Character

(char_then_puppet)

Printable Then the Action Dog illustration on a popsicle stick.



Then the Action Dog

PRINT SPECS

Size: letter portrait

Colour: bw

Pages per set: 1

Laminate: yes

USE IN CLASS

Quantity: 1 puppet

Lessons: 2, 3, 4, 5

Prep: ~10 min

TEACHER PREP

1. Print on cardstock.

2. Cut.

3. Glue popsicle stick.

4. Optional: laminate.

5. Store flat.

Name: _____

Date: _____

Summative Assessment Rubric

Expectation	Level 1 (Beginning)	Level 2 (Developing)	Level 3 (Achieving)	Level 4 (Exceeding)
C3.1 solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves conditional statements and other control structures	Cannot write IF-THEN code.	Writes one branch of IF-THEN with prompts.	Writes complete two-branch IF-THEN code for a math task.	Writes 3-branch IF-THEN-ELSE-IF code; explains the structure.
C3.2 read and alter existing code, including code that involves conditional statements and other control structures, and describe how changes to the code affect the outcomes	Cannot read or alter IF-THEN code.	Traces simple code with prompts.	Traces and predicts outcomes; alters code and predicts new outcomes.	Identifies which inputs change after alteration; describes boundary cases.

Junior Conditional Coder Certificate

This certificate is proudly presented to:

For successfully completing the 5-Day If-Then Detectives unit and demonstrating the skills of a Junior Conditional Coder. Inspector If and Then the Action Dog congratulate the student.

By completing this unit, this student has demonstrated:

- Reading IF-THEN statements
- Writing two-branch IF-THEN code
- Tracing code with sample inputs
- Altering code and predicting effects
- Inventing and tracing a capstone IF-THEN project

Marketplace Listing

A 5-day Ontario Grade-2 unit on coding without screens (C3.1-C3.2) framed around 'If-Then Detectives' — Inspector If the Conditional Cat and her partner Then the Action Dog.

Price: \$9.99 CAD

Pages: ~32

Classroom time: 250 min

Teacher prep: ~90 min

Top tags: grade-2, math, ontario-curriculum, coding, algebra, conditionals, without-screens, canadian

What's included:

- 5 fully-scripted lesson plans with 50-minute durations and 250 minutes of total teaching time
 - 5 student worksheets aligned to each lesson
 - 1 mid-unit formative assessment (IF-THEN Check-In)
 - 10 printable teacher manipulatives plus 2 character puppets: IF-THEN statement cards (24), condition cards (12), action cards (12), input number cards (20),...
 - 1 end-of-unit reflection sheet with 5 self-assessment prompts
 - 4 class observation trackers: 1 diagnostic + 3 formative
 - 1 two-row summative rubric covering C3.1-C3.2 across 4 levels
 - 1 summative-administration guide tying the rubric to Lesson 5 capstone
 - Junior Conditional Coder certificate (one per child)
 - Detailed expanded walkthroughs covering common misconceptions
 - Differentiation tracks built into every lesson